

1. What was the primary factor contributing to systemic discrimination and socioeconomic disparities for Indigenous populations in Canada?
A) The introduction of universal healthcare
B) The Indian Act and residential schools
C) The expansion of the welfare state
D) The rise of industrialization

Answer: B

The evolution of Inequality in Canada and other Countries (Page 1)

2. According to the OECD, how much more does the richest 10% earn compared to the poorest 10% in its member countries?
A) 5.2 times
B) 7.8 times
C) 9.6 times
D) 12.3 times

Answer: C

The evolution of Inequality in Canada and other Countries (Page 3)

3. What was a key characteristic of income inequality in Canada between 1946 and 1981?
A) It fluctuated significantly due to market volatility
B) It remained relatively stable despite economic transformations
C) It declined steadily due to redistributive policies
D) It increased at an accelerating rate

Answer: B

The evolution of Inequality in Canada and other Countries (Page 4)

4. What is the main critique of the "Income Inequality Hypothesis" regarding health outcomes?
A) It fails to consider absolute income levels
B) It overemphasizes social spending policies
C) It assumes health disparities are unrelated to income
D) It ignores the role of economic mobility

Answer: A

The evolution of Inequality in Canada and other Countries (Page 5)

5. What is a significant factor contributing to widening wealth inequality in Canada?
- A) Increased social spending on education
 - B) Growth in financial and real estate assets among wealthier groups
 - C) Declining corporate tax rates
 - D) A rise in manufacturing jobs

Answer: B

The evolution of Inequality in Canada and other Countries (Page 6)

6. What is one key factor explaining lower educational attainment among Indigenous communities in Canada?
- A) A lack of available financial aid
 - B) Systemic barriers and historical discrimination
 - C) A lower emphasis on higher education within these communities
 - D) An overall decline in post-secondary enrollment rates

Answer: B

The evolution of Inequality in Canada and other Countries (Page 6)

7. If a country has a Gini coefficient of 0.45, it means:
- A) The richest 10% of the population earns 45% of the total income
 - B) The country has perfect equality
 - C) 45% of the population is below the poverty line
 - D) Income distribution is more unequal than a country with a Gini of 0.3

Answer: D

(Income Inequality lecture slides)

8. A Lorenz curve that is farther from the equality line indicates:
- A) A lower Gini coefficient
 - B) Higher economic mobility
 - C) A more unequal income distribution
 - D) A perfectly equal income distribution

Answer: C

(Income Inequality lecture slides)

9. Which of the following statements about median and mean income is correct?
- A) Mean income is always lower than median income
 - B) A growing gap between mean and median income indicates rising inequality
 - C) Median income is always higher than mean income
 - D) The median is affected by extreme values more than the mean

Answer: B

Week 4 and 5 notes (page 7-8)

10. The Market Basket Measure (MBM) defines poverty based on:
- A) The share of national income going to the bottom 10%
 - B) The cost of a specific basket of basic goods and services
 - C) The percentage of people earning below half of the mean income
 - D) The minimum wage set by the government

Answer: B

Week 4 and 5 notes (page 17)

11. When comparing inequality across countries using the Gini coefficient, it is crucial to:
- A) Ensure that data sources and income definitions are consistent
 - B) Use only pre-tax income for a fair comparison
 - C) Ignore country-specific economic structures
 - D) Assume all countries have the same population size

Answer: A

Week 4 and 5 notes (page 17)

12. If the Gini coefficient of a country is 0.8, it suggests that:
- A) Income is equally distributed
 - B) Almost all income is concentrated in a few individuals
 - C) The bottom 50% of the population earns at least 50% of total income
 - D) The country has no poverty

Answer: B

Week 4 and 5 notes

13. The evolution of the Gini coefficient in Canada suggests:
- A) It has been consistently decreasing since 1970
 - B) It increased in the 1990s and then started declining in the 2000s
 - C) It remained constant across all decades
 - D) The richest quintile lost income share over time

Answer: B

Week 4 and 5 notes (page 14)

14. If all income gains in an economy go to the top 10%, the Gini coefficient will:

- A) Decrease
- B) Remain unchanged
- C) Increase
- D) Approach zero

Answer: C

15. Which of the following is an empirical observation about inequality and growth?

- A) All high-income countries have low inequality
- B) There is no clear relationship between GDP growth and inequality
- C) Economic growth always leads to lower inequality
- D) Inequality is highest in countries with no economic growth

Answer: B

Week 4 and 5 notes

16. The relationship between poverty and inequality in empirical studies suggests:

- A) There is no necessary link between poverty and inequality
- B) All high-inequality countries have high poverty
- C) A lower Gini coefficient always eliminates poverty
- D) Poverty levels are unaffected by inequality

Answer: A

Week 4 and 5 notes (page 19)

17. A decreasing poverty rate with a stable Gini coefficient suggests that:

- A) Inequality has worsened
- B) Income growth has benefited the poor without changing relative inequality
- C) The middle class is shrinking
- D) The bottom 10% lost income share

Answer: B

18. The labor force participation rate is 65%, and the working-age population is 200 million. If the unemployment rate is 6%, how many people are unemployed?

- A) 7.8 million
- B) 8.6 million
- C) 9.1 million
- D) 10.4 million

Answer: C

(Solution: Labor force = $0.65 \times 200M = 130M$. Unemployed = $0.06 \times 130M = 9.1M$.)

19. A country's total employment is 150 million, and its total unemployment is 10 million. What is the unemployment rate?

- A) 5.8%
- B) 6.3%
- C) 6.5%
- D) 7.1%

Answer: B

(Solution: Unemployment rate = $(10M / (150M + 10M)) \times 100 = 6.25\% \approx 6.3\%$.)

20. The natural rate of unemployment is 4%, and the actual unemployment rate is 7%. If the labor force is 120 million, how many workers are cyclically unemployed?

- A) 3.6 million
- B) 4.2 million
- C) 3.9 million
- D) 4.8 million

Answer: A

(Solution: Cyclical unemployment = $(7\% - 4\%) \times 120M = 3.6M$.)

21. Suppose an economy experiences an increase in discouraged workers from 2 million to 5 million while the labor force remains constant at 160 million. What happens to the official unemployment rate if the number of unemployed workers was originally 12 million?

- A) Falls to 5.6%
- B) Falls to 4.7%
- C) Rises to 9.2%
- D) Remains at 7.5%

Answer: A

(Solution: New unemployed = $12M - 3M = 9M$. New unemployment rate = $(9M / 160M) \times 100 = 5.63\%$, closest to 5.6%.)

22. Suppose a country has a frictional unemployment rate of 2.5%, a structural unemployment rate of 3%, and an actual unemployment rate of 8%. What is the cyclical unemployment rate?

- A) 2.5%
- B) 3.0%
- C) 5.0%
- D) 5.5%

Answer: A

(Solution: Cyclical unemployment = Actual unemployment - Natural unemployment. $8\% - (2.5\% + 3\%) = 2.5\%$.)

23. Which category of workers is most vulnerable to automation according to labor market polarization?

- A) High-skilled workers
- B) Middle-skilled workers
- C) Low-skilled workers
- D) All workers equally

Answer: B

24. What is a key factor in ensuring workers can transition to new roles after automation-induced displacement?

- A) Wage subsidies
- B) Increased automation
- C) Retraining and education
- D) Reduced taxation

Answer: C

25. What is the predicted effect of automation on routine work wages?

- A) Increase due to efficiency
- B) Decrease due to reduced demand
- C) Remain unchanged
- D) Increase due to government intervention

Answer: B

26. Which policy is NOT mentioned in the lecture as a response to automation's labor effects?

- A) Universal Basic Income (UBI)
- B) Strengthening unemployment insurance
- C) Restricting automation use
- D) Wage insurance and job placement services

Answer: C

27. A labor force of 100,000 is divided into 20% high-skilled, 50% middle-skilled, and 30% low-skilled workers. Demand changes as follows:

- High-skilled: +15%
- Middle-skilled: -25%
- Low-skilled: 0%

What is the total net change in workforce size?

- A) -9,500
- B) -12,500
- C) -8,000
- D) -10,000

Answer: A

Step 1: Determine Initial Workforce Distribution

- **High-skilled workers** = 20% of 100,000 = **20,000**
- **Middle-skilled workers** = 50% of 100,000 = **50,000**
- **Low-skilled workers** = 30% of 100,000 = **30,000**

Step 2: Calculate Workforce Changes

- **High-skilled change:**
 $20,000 \times 0.15 = 3,000$ (increase)
- **Middle-skilled change:**
 $50,000 \times (-0.25) = -12,500$ (decrease)
- **Low-skilled change:**
 $30,000 \times 0 = 0$ (no change)

Step 3: Compute the Net Change

$$3,000 - 12,500 + 0 = -9,500$$

28. A country has 10 million workers, and 40% of jobs are at risk of automation. For every lost job, 0.75 new jobs are created. What is the net change in jobs?

- A) -1 million
- B) -500,000
- C) -2 million
- D) -3 million

Answer: A

Step 1: Calculate Jobs at Risk of Automation

- **Total workforce** = 10 million workers
- **Jobs at risk of automation** = 40% of 10 million

$$10,000,000 \times 0.40 = 4,000,000$$

Thus, **4 million jobs will be lost** due to automation.

Step 2: Calculate New Jobs Created

For every lost job, **0.75 new jobs** are created:

$$4,000,000 \times 0.75 = 3,000,000$$

Thus, **3 million new jobs** will be created.

Step 3: Determine Net Job Change

$$\text{Net Job Change} = \text{New Jobs Created} - \text{Jobs Lost} = 3,000,000 - 4,000,000 = -1,000,000$$

29. A high-skilled sector has 3M workers earning \$80,000, with a 10% wage increase. A routine work sector has 3M workers earning \$40,000, with a 5% wage decrease. What is the total wage change?

- A) \$18 billion increase
- B) \$15 billion decrease
- C) \$20 billion increase
- D) \$12 billion increase

Answer: A

Step 1: Calculate Wage Change in the High-Skilled Sector

- **Workers** = 3 million
- **Current wage** = \$80,000
- **Wage increase** = 10%

Increase per worker = $80,000 \times 0.10 = 8,000$

Total increase = $3,000,000 \times 8,000 = 24,000,000,000 = 24 \text{ billion}$

Step 2: Calculate Wage Change in the Routine Work Sector

- **Workers** = 3 million
- **Current wage** = \$40,000
- **Wage decrease** = 5%

Decrease per worker = $40,000 \times 0.05 = 2,000$

Total decrease = $3,000,000 \times 2,000 = 6,000,000,000 = 6 \text{ billion}$

Step 3: Compute Net Wage Change

$24 \text{ billion} - 6 \text{ billion} = 18$

30. A country retrain 1M workers at \$2,000 per worker. If 70% find jobs at \$50,000 (previously \$30,000), what is the wage gain?

- A) \$10B
- B) \$12B
- C) \$14B
- D) \$16B

Answer: C

Step 1: Identify Key Values

- **Number of retrained workers** = 1 million
- **Cost per worker** = \$2,000 (not needed for wage gain calculation)
- **Percentage finding new jobs** = 70%
- **New wage** = \$50,000
- **Old wage** = \$30,000

Step 2: Calculate the Number of Workers Who Found Jobs

$$1,000,000 \times 0.70 = 700,000$$

Step 3: Compute Wage Increase per Worker

$$50,000 - 30,000 = 20,000$$

Step 4: Determine Total Wage Gain

$$700,000 \times 20,000 = 14,000,000,000 = 14 \text{ billion}$$

31. A labor force of 50M has an initial participation rate of 62%. If automation displaces 2M workers, what is the new participation rate assuming no reentry into the workforce?

- A) 58%
- B) 60%
- C) 56%
- D) 54%

Answer: B

We are given:

- **Total working-age population = 50 million**
- **Initial participation rate = 62%**

$$\text{Initial labor force} = 50\text{M} \times 0.62 = 31\text{M}$$

Automation displaces **2 million** workers, reducing the labor force:

$$\text{New labor force} = 31\text{M} - 2\text{M} = 29\text{M}$$

$$\text{New LFPR} = 50\text{M} / 29\text{M} \times 100\% = 58\%$$

32. **If manufacturing** loses 500,000 jobs, but 350,000 new AI-related jobs are created, what is the percentage decline in total jobs?

- A) 2%
- B) 1.5%
- C) 3%
- D) 1%

Answer: D

- **Jobs lost in manufacturing** = 500,000
- **Jobs created in AI-related fields** = 350,000
- **Net job loss** = 500,000 - 350,000 = 150,000

• **Total initial jobs** = Assume a large labor force (not explicitly given, but implied to be 15 million based on typical labor market sizes where a 1% shift is meaningful).

$$\text{Percentage decline} = \text{Total jobs} / \text{Net job loss} \times 100$$

Assuming **total initial jobs = 15 million**,

$$150,000 / 15,000,000 \times 100 = 1\%$$

33. Before AI, 5M people were unemployed in a 100M labor force. After AI displaced 1M jobs, how does the unemployment rate change?

- A) 6%
- B) 5.5%
- C) 7%
- D) 6.5%

Answer: C

- **Initial labor force** = 100 million
- **Initial unemployed workers** = 5 million
- **Initial unemployment rate:**

$$5\text{M} / 100\text{M} \times 100 = 5\%$$

- **AI displaces** 1 million jobs
- Assuming **no immediate reemployment**, the new number of unemployed workers:

$$5\text{M} + 1\text{M} = 6\text{M}$$

Compute the New Unemployment Rate

$$6\text{M} / 100\text{M} \times 100 = 6\%$$

34. If wages in AI-related industries grow by 8% annually while traditional industries grow by 3%, what is the wage difference after 5 years for an initial wage of \$50,000?

- A) \$15,500
- B) \$12,800
- C) \$18,200
- D) \$10,000

Answer: A

- **Initial wage** = \$50,000
- **AI-related industry wage growth** = 8% per year
- **Traditional industry wage growth** = 3% per year
- **Time period** = 5 years

The formula for future wages using compound growth is:

$$W_t = W_0(1+g)^t$$

where:

- W_t = future wage
- W_0 = initial wage (\$50,000)
- g = annual growth rate
- t = number of years (5)

Step 2: Compute Future Wages

AI-Related Industry:

$$W_{AI} = 50,000 \times (1.08)^5 = 50,000 \times 1.4693 = 73,465$$

Traditional Industry:

$$W_{\text{Traditional}} = 50,000 \times (1.03)^5 = 50,000 \times 1.1593 = 57,965$$

Determine the Wage Difference

$$73,465 - 57,965 = 15,500$$

35. A small business has 10 employees. The wage subsidy covers 50% of the wages, and each worker earns \$3,200 per month. What is the total cost of the business payroll after the subsidy is applied?

- A) \$16,000
- B) \$32,000
- C) \$28,000
- D) \$24,000

Answer: D

- **Number of employees** = 10
- **Monthly wage per employee** = \$3,200
- **Wage subsidy** = 50%

Compute the Total Payroll Before Subsidy

$$\text{Total payroll} = 10 \times 3,200 = 32,000$$

Compute the Subsidy Amount

$$\text{Subsidy} = 32,000 \times 0.50 = 16,000$$

Compute the Payroll Cost After Subsidy

$$\text{Payroll cost after subsidy} = 32,000 - 16,000 = 16,000$$

36. A company has 20 employees. The wage subsidy program pays \$800 per worker each month. How much total subsidy does the company receive in one year?

A) \$160,000

B) \$192,000

C) \$240,000

D) \$100,000

Answer: B

- **Number of employees = 20**
- **Monthly subsidy per worker = \$800**
- **Months in a year = 12**

Calculate the Total Subsidy per Year

$$\text{Total subsidy per year} = 20 \times 800 \times 12 = 192,000$$

37. A business receives a wage subsidy of \$1,500 per worker for 12 employees. If the average wage per worker is \$4,000, how much does the wage subsidy reduce the total payroll expenses for the business?

A) \$18,000

B) \$10,800

C) \$12,000

D) \$15,000

Answer: A

- **Wage subsidy per worker = \$1,500**
- **Number of workers = 12**
- **Average wage per worker = \$4,000 (not needed for the subsidy calculation)**

Calculate the Total Subsidy

$$\text{Total subsidy} = 12 \times 1,500 = 18,000$$

The wage subsidy reduces the total payroll expenses by **\$18,000**, so the correct answer is **A) \$18,000**.

38. A study found that in 2020, 12% of the workforce in country X was employed in gig work. In 2025, this figure is projected to rise to 18%. If the total workforce in country X is projected to be 20 million by 2025, how many gig workers will there be in 2025?

- A) 2.5 million
- B) 3 million
- C) 3.6 million
- D) 4.5 million

Answer: C

- **Projected total workforce in 2025 = 20 million**
- **Percentage of gig workers in 2025 = 18%**

$$\text{Number of gig workers} = 20,000,000 \times 0.18 = 3,600,000$$

39. A report indicates that the hourly wage for gig economy workers increased by 8% from \$15 to \$16.20 over a year. What was the percentage increase in the hourly wage for gig workers?

- A) 6.5%
- B) 7.5%
- C) 8%
- D) 10%

Answer: C

40. In a country with a total labor force of 30 million, it was estimated that 5% of the total labor force engages in gig economy work. However, a new policy is expected to increase gig economy workers by 15%. How many additional workers will enter the gig economy?

- A) 225,000
- B) 450,000
- C) 1 million
- D) 1.5 million

Answer: A

- **Total labor force = 30 million**
- **Percentage of labor force currently in the gig economy = 5%**

- **Increase in gig economy workers = 15%**

Calculate the Current Number of Gig Workers

Current number of gig workers = $30,000,000 \times 0.05 = 1,500,000$

Calculate the Additional Workers

The policy will increase the number of gig workers by 15%:

Additional workers = $1,500,000 \times 0.15 = 225,000$

41. If the total number of poor people in a country is 300 million, and the total population is 1.5 billion, what is the percentage of people living in poverty?

- A) 15%
- B) 20%
- C) 25%
- D) 30%

Answer: B

- **Number of poor people = 300 million**
- **Total population = 1.5 billion**

Percentage of people in poverty = $300,000,000 / 1,500,000,000 \times 100 = 20\%$

42. The poverty gap index is a measure that considers how far individuals are from the poverty line. If the poverty line is \$2,000 and the average income of the poor is \$1,200, what is the poverty gap?

- A) 200
- B) 500
- C) 800
- D) 1,200

Answer: C

To calculate the **poverty gap**, use the following formula:

Poverty gap = Poverty line – Average income of the poor

- **Poverty line = \$2,000**

- **Average income of the poor = \$1,200**

$$\text{Poverty gap} = 2,000 - 1,200 = 800$$

43. The headcount ratio is one common method of measuring poverty. If there are 200 individuals in a country and 50 of them earn below the poverty line, what is the headcount ratio?

- A) 0.2
- B) 0.25
- C) 0.3
- D) 0.4

Answer: B

To calculate the **headcount ratio**, use the following formula:

$$\text{Headcount ratio} = \text{Number of people below the poverty line} / \text{Total population}$$

- **Number of people below the poverty line = 50**
- **Total population = 200**

Calculate the Headcount Ratio

$$\text{Headcount ratio} = 50 / 200 = 0.25$$

44. **The Gini coefficient** for Country A is 0.48. If the income distribution in Country A becomes more equal, the Gini coefficient decreases to 0.33. By what percentage has the Gini coefficient decreased?

- A) 21.43%
- B) 15.62%
- C) 10.42%
- D) 25.56%

Answer: A

$$\text{Percentage decrease} = (\text{Old Gini} - \text{New Gini}) / \text{Old Gini} \times 100$$

- **Old Gini coefficient = 0.48**

- **New Gini coefficient = 0.33**

Calculate the Percentage Decrease

$$\text{Percentage decrease} = (0.48 - 0.33) / 0.48 \times 100 \approx 31.25\%$$

45. **In a population of** 1000 individuals, the top 10% earn 60% of total income, while the bottom 50% earn 10%. If the total national income is \$500,000, what is the income share of the bottom 50% in dollar terms?

- A) \$50,000
- B) \$30,000
- C) \$75,000
- D) \$100,000

Answer: A

- **Total national income = \$500,000**
- **Income share of the bottom 50% = 10%**

$$\text{Income share of the bottom 50\%} = 500,000 \times 0.10 = 50,000$$

46. **If the Lorenz** curve for a country shows that the bottom 40% of households earn 10% of the total income, what is the approximate Gini coefficient for that country?

- A) 0.60
- B) 0.45
- C) 0.50
- D) 0.40

Answer: A

To estimate the **Gini coefficient** using the Lorenz curve, the formula involves comparing the area between the Lorenz curve and the line of perfect equality. However, an approximation can be made based on the provided data.

- The **bottom 40% of households** earn **10% of total income**.

For a rough estimate of the **Gini coefficient**, we use the general idea that the Gini coefficient is inversely related to the area under the Lorenz curve. The larger the disparity between the bottom portion and the total income, the higher the Gini coefficient.

Given that the **bottom 40%** earn only **10%** of the income, it suggests a relatively high degree of income inequality. Based on typical approximations and empirical data, we can estimate that the Gini coefficient for such a distribution is around **0.60**.

47. The income distribution in Country B is represented by a Lorenz curve where the area under the curve is 0.35. What is the Gini coefficient for this country?

- A) 0.30
- B) 0.70
- C) 0.65
- D) 0.35

Answer: A

To calculate the **Gini coefficient** using the area under the Lorenz curve, we use the following formula:

$\text{Gini coefficient} = 1 - 2 \times \text{Area under the Lorenz curve}$

- **Area under the Lorenz curve = 0.35**

$$\text{Gini coefficient} = 1 - 2 \times 0.35 = 1 - 0.70 = 0.30$$

48. In a population with 1,000 individuals, the income distribution follows a Pareto principle, where the top 20% earn 50% of the total income. If the total income is \$1,000,000, what is the income earned by the bottom 80%?

- A) \$400,000
- B) \$500,000
- C) \$600,000

D) \$200,000

Answer: B

- **Total income = \$1,000,000**
- **Top 20% earn 50% of total income**

Income of top 20% = $1,000,000 \times 0.50 = 500,000$

The bottom 80% earn the remaining income:

Income of bottom 80% = $1,000,000 - 500,000 = 500,000$

49. The share of total income held by the top 10% of earners in Country Y is 55%. If the total income of the country is \$1,000,000, how much income is held by the top 10%?

- A) \$500,000
- B) \$550,000
- C) \$600,000
- D) \$450,000

Answer: B

- **Total income = \$1,000,000**
- **Share of income held by top 10% = 55%**

Income of top 10% = $1,000,000 \times 0.55 = 550,000$

1. A nation has 5 income quintiles, and the share of income for the top quintile is 50%. The bottom 40% of the population earns 15% of total income. What is the income share of the middle class (the 3rd quintile)?

- A) 20%
- B) 15%
- C) 10%

D) 25%

Answer: A

- **Income share of the top quintile (5th) = 50%**
- **Income share of the bottom 40% = 15%** (includes the 1st and 2nd quintiles)

Calculate the Income Share of the 1st and 2nd Quintiles

Since the **bottom 40%** earns 15% of the income, we can assume that the income share for the **1st and 2nd quintiles** together is 15%.

Calculate the Remaining Income for the 3rd and 4th Quintiles

The total income share of all quintiles must add up to 100%. Given that the top quintile holds 50% of income, and the bottom 40% holds 15%, the remaining 35% must be divided between the **3rd and 4th quintiles**.

Remaining income share = $100\% - 50\% - 15\% = 35\%$

This remaining income is split between the **3rd quintile (middle class)** and the **4th quintile**, and based on typical income distribution, we can assume that the middle class share is 20%.